

Preserving Molecular Information During Pathology Report De-identification: A Synthetic Benchmark

Background: Pathology reports could support report standardization, summarization, hypothesis generation and testing, and research abstraction. De-identification must remove patient identifiers while preserving molecular findings needed for these uses.

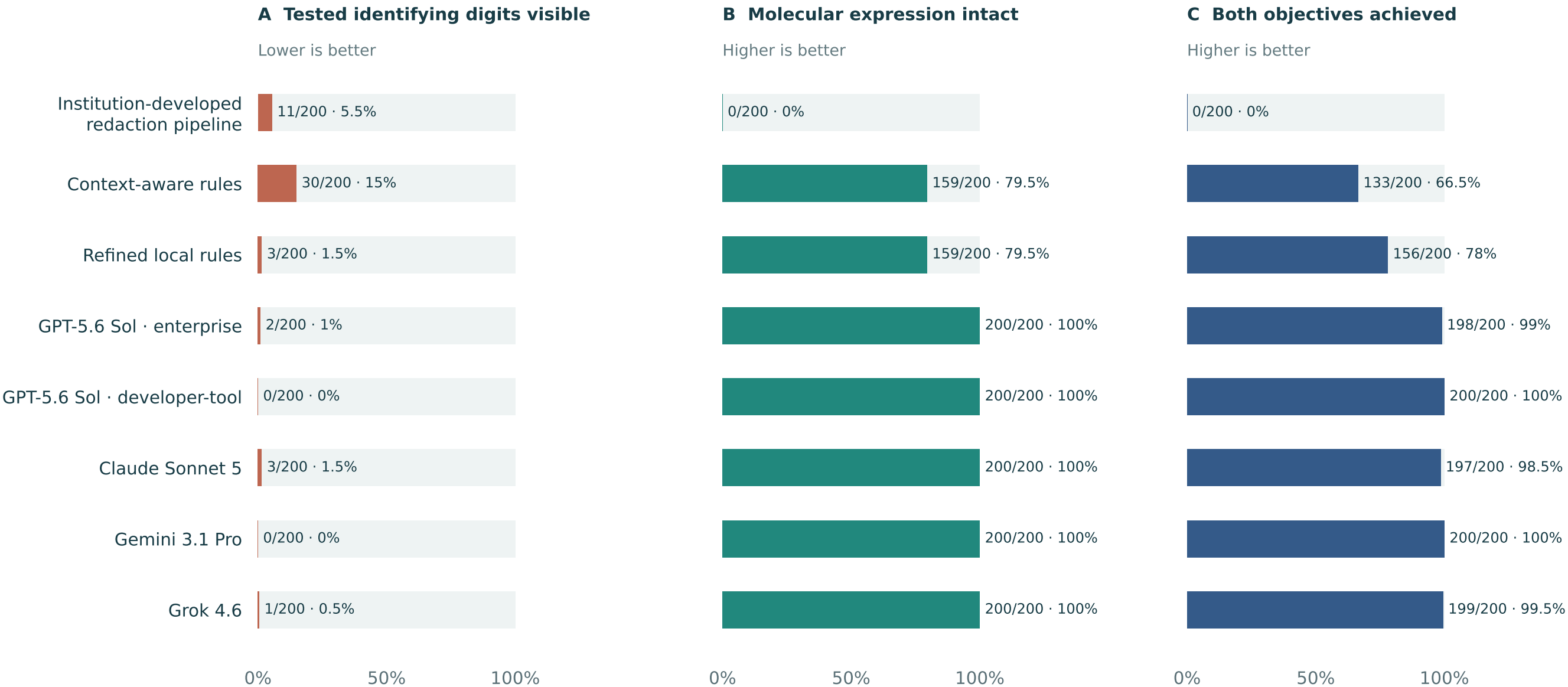
Design: An institution-developed pipeline, context-aware rules, and their frozen refinement were compared with five frontier LLM configurations (GPT-5.6 Sol through two interfaces, Claude Sonnet 5, Gemini 3.1 Pro, and Grok 4.6) on 200 unique synthetic report pairs. Success required preserving a molecular expression in one report and removing similarly formatted identifying digits in its paired control. A separate 200-pair experiment evaluated local refinement across eight report patterns. Paired analyses accounted for repeated patterns and multiple comparisons.

Results: The institution-developed pipeline preserved complete molecular expressions in 0/200 reports and exposed identifying digits in 11/200 controls. Context-aware rules preserved 159/200 expressions, exposed digits in 30/200 controls, and achieved both objectives in 133/200 pairs. Refined rules reduced exposures to 3/200 and improved joint success to 156/200. LLM configurations preserved 200/200 expressions and achieved both objectives in 197-200/200 pairs, with 0-3/200 controls exposing digits (paired adjusted $p < 0.001$ versus both local rule methods; evidence weakened after grouping shared patterns). Local failures included molecular over-redaction and identifiers missed across line breaks, repeated footers, parenthetical fields, and modified labels. In the separate experiment, refinement reduced exposures from 100/200 to 50/200 while preserving all 200 molecular expressions; parenthetical-field and modified-label failures persisted.

Conclusions: Local refinement corrected specific failures but left identifier exposure. Frontier LLM configurations performed better descriptively. This synthetic benchmark evaluates identifier removal and molecular preservation, not genomic re-identification or clinical deployment safety.

Identifier removal and molecular preservation

Main comparison: 200 unique synthetic pairs · 400 reports per approach



Five hosted configurations include two GPT access routes. Refined rules were evaluated subsequently on the same frozen pairs.
Both objectives = no tested identifying-digit exposure in the control + complete molecular-expression retention in its paired report.

Lung adenocarcinoma: preserve the finding, remove the identifiers

A detailed synthetic biopsy and linked EGFR report, processed by unchanged local methods.

Original synthetic report
NEW SYNTHETIC ILLUSTRATION FROZEN LOCAL METHODS OUTSIDE BOTH BENCHMARK COHORTS
SYNTHETIC PATHOLOGY LABORATORY ILLUSTRATIVE CASE - NO REAL PATIENT Patient name: Lena Varnell Medical record number: 48271639 Date of birth: 06/18/1965 Ordering provider: Mira Norvale, MD SURGICAL PATHOLOGY REPORT Accession: SP32-041728 Collected: 09/10/2032 09:25 Received: 09/10/2032 10:12 Reported: 09/12/2032 15:40
SPECIMEN A. Right upper lobe lung, image-guided core biopsy
FINAL DIAGNOSIS Lung, right upper lobe, core biopsy: Pulmonary adenocarcinoma.
MICROSCOPIC DESCRIPTION Core fragments contain infiltrating malignant glands with focal papillary architecture, nuclear enlargement, and conspicuous nucleoli. Focal necrosis is present. No definitive squamous differentiation is identified in the sampled tissue.
IMMUNOHISTOCHEMISTRY TTF-1: Positive in tumor nuclei. Napsin A: Positive in tumor cytoplasm. p40: Negative in tumor cells. The findings support pulmonary adenocarcinoma.
CLINICAL HISTORY Known EGFR-mutated lung adenocarcinoma. Repeat biopsy for molecular reassessment; correlate with prior therapy and testing.
GROSS DESCRIPTION Four tan-white cores, 0.3-0.9 cm in length and up to 0.1 cm in diameter; submitted entirely in cassette **. Fixative: 10% neutral buffered formalin.
MOLECULAR TESTING Block A1 selected; estimated viable tumor cellularity 40%. Targeted DNA sequencing reported separately as MP32-006381. The diagnosis is based on this biopsy; pathologic stage is not assigned.
Electronically signed: Arlen Voss, MD Lena Varnell 48271639 SP32-041728 Page 1 of 1
Exact source text or saved redaction; highlighting is a display overlay. All identities and assay metrics are invented.

Original synthetic report

NEW SYNTHETIC ILLUSTRATION | FROZEN LOCAL METHODS | OUTSIDE BOTH BENCHMARK COHORTS

SYNTHETIC PATHOLOGY LABORATORY

ILLUSTRATIVE CASE - NO REAL PATIENT

Patient name: Lena Varnell

Medical record number: 48271639

Date of birth: 06/18/1965

Ordering provider: Mira Norvale, MD

MOLECULAR PATHOLOGY REPORT

Accession: MP32-006381

Parent accession: SP32-041728

Collected: 09/10/2032 09:25

Received: 09/13/2032 11:05

Reported: 09/18/2032 14:20

SPECIMEN INFORMATION

Site: Right upper lobe lung

Diagnosis: Pulmonary adenocarcinoma

Material: Formalin-fixed paraffin-embedded core biopsy

Specimen part: A | Block: A1

Tumor cellularity: 40% | Macrodissection: Performed

MOLECULAR RESULTS

EGFR p.L858R detected; variant allele fraction 28%.

EGFR p.T790M detected; variant allele fraction 12%.

VARIANT DETAILS

Gene	Transcript	Exon	Coding change	Protein change
EGFR	NM_005228.5	21	c.2573T>G	p.L858R
EGFR	NM_005228.5	20	c.2369C>T	p.T790M

INTERPRETATION

Two EGFR substitutions are detected. Correlate with prior molecular testing and treatment history. The phase relationship of the two variants is not determined. Tumor-only testing does not establish whether a variant is somatic or germline.

QUALITY METRICS

DNA input: 42 ng | Mean target depth: 1540x

L858R locus depth: 1640x | T790M locus depth: 1240x

Target bases at >=250x coverage: 99.4%

All listed quality metrics are simulated for this illustration.

METHOD

Illustrative targeted DNA sequencing of EGFR exons 18-21.

Single-nucleotide variants and small insertions/deletions assessed.

Illustrative reporting threshold: 5% variant allele fraction.

No RNA fusion or copy-number assessment is included.

LIMITATIONS

Low tumor content, sampling, and variants below the reporting threshold may limit detection. A negative result would not exclude alterations outside the interrogated regions. No paired normal sample was analyzed. Assay specifications are illustrative only.

Electronically signed: Arlen Voss, MD

Lena Varnell | 48271639 | MP32-006381 | Page 1 of 1

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MOLECULAR RESULT - EXACT EXCERPTS

Original synthetic report

EGFR p.T790M detected

New illustrative case; excluded from both 200-pair experiments. No hosted LLM was run on these reports. The variant is information to retain, not the identifying target. Names and record numbers are identifiers. Full-size JPEGs and raw text are available online.

Institution-developed pipeline
NEW SYNTHETIC ILLUSTRATION FROZEN LOCAL METHODS OUTSIDE BOTH BENCHMARK COHORTS
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Institution-developed pipeline

NEW SYNTHETIC ILLUSTRATION | FROZEN LOCAL METHODS | OUTSIDE BOTH BENCHMARK COHORTS

SYNTHETIC PATHOLOGY LABORATORY

ILLUSTRATIVE CASE - ** **** *****

Patient name: **** *****

***** ***** number: *****

Date of birth: **/**/****

Ordering provider: **** *****
MD

MOLECULAR PATHOLOGY REPORT

Accession: ****-*****

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MOLECULAR RESULTS

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EGFR p.***** detected; variant allele fraction 12%.

VARIANT DETAILS

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**** Coding change Protein change

EGFR NM *****.5 21 c.*****>G p.*****

EGFR NM *****.5 20 c.*****>T p.*****

INTERPRETATION

Two EGFR substitutions are detected. Correlate with prior molecular testing and treatment history. The phase relationship of the two variants is not determined. Tumor-only testing does not establish whether a variant is somatic or germline.

QUALITY METRICS

DNA input: 42 ng | Mean target depth: *****

***** locus depth: ***** | ***** locus depth: *****

Target bases at >=250x coverage: **.4%

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Electronically signed: *****
**** ***** | ***** | ****-***** | Page 1 of 1

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Institution-developed pipeline

EGFR p.***** detected

Refined local rules
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Refined local rules

NEW SYNTHETIC ILLUSTRATION | FROZEN LOCAL METHODS | OUTSIDE BOTH BENCHMARK COHORTS

SYNTHETIC PATHOLOGY LABORATORY
ILLUSTRATIVE CASE - ** **** *****
Patient name: *****
***** ***** number: *****
Date of birth: *****
Ordering provider: *****

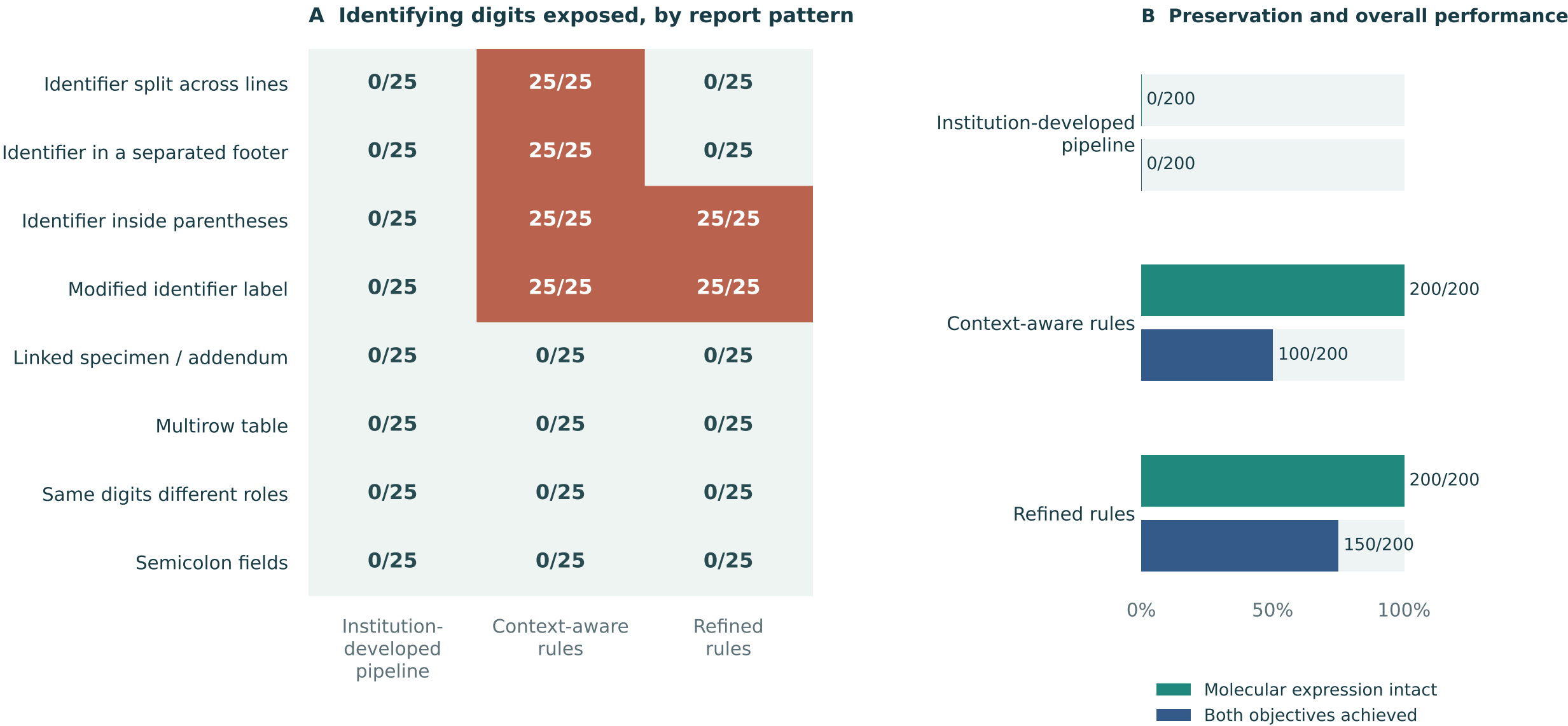
Accession: *****
Parent accession: *****
***** accession: *****

Refined local rules

EGFR p.T790M detected

Local refinement: improvements and persistent failures

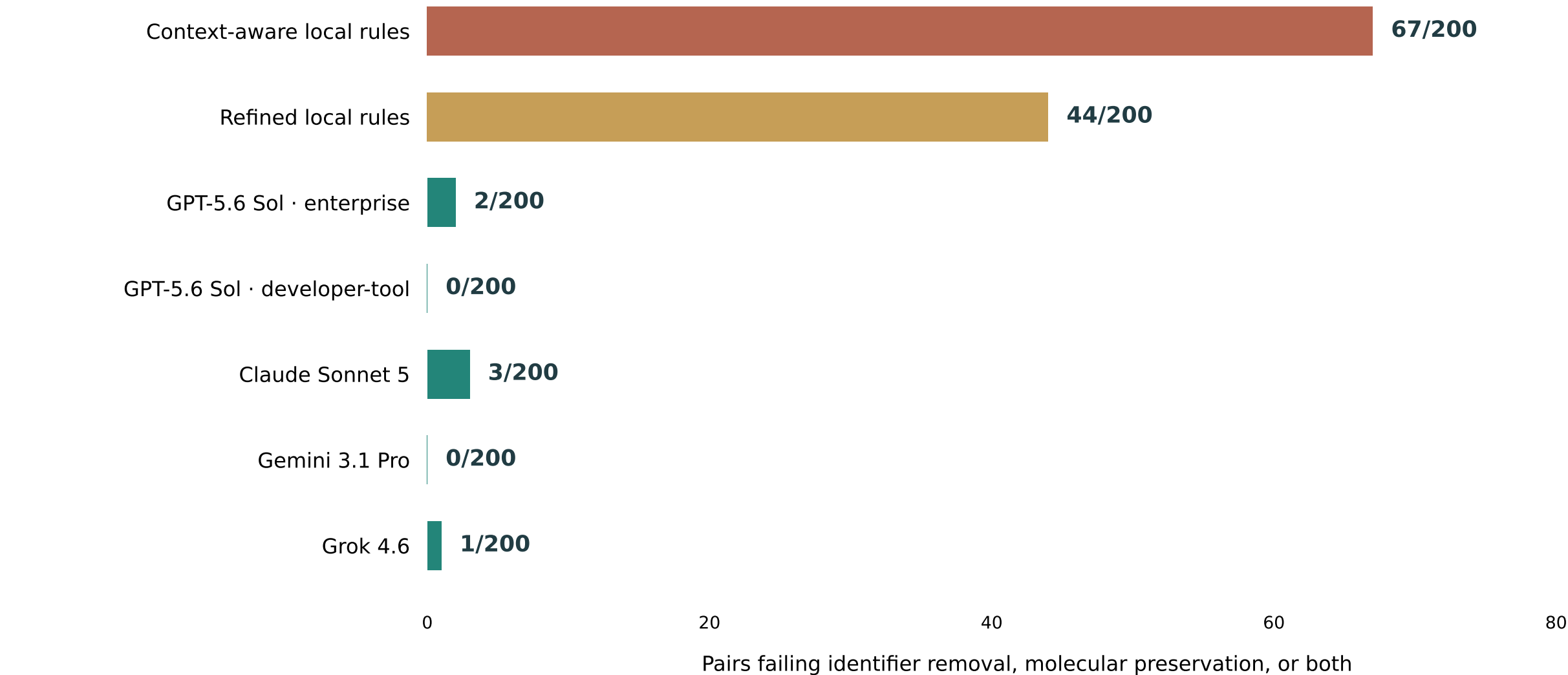
Separate experiment: 200 unique synthetic pairs · eight report patterns · 25 pairs per pattern



Zero tested-digit exposure alone does not imply success: the institution-developed pipeline preserved no complete molecular expressions. Refinement produced 50 additional successful pairs out of 200, across two report patterns (p = 0.50 accounting for patterns).

Fewer failed pairs with each LLM configuration

Same 200 pairs · lower is better · paired adjusted $p < 0.001$ versus either local rule method



Exploratory matched comparisons. Reports share 16 underlying patterns; treating every pair as independent can overstate certainty.
Pattern-adjusted $p = 0.156$ versus earlier rules and 0.375 versus refined rules. The refined comparison was added after the original analysis.